

Figure 5: Snort in network intrusion mode

If you need to enable network intrusion mode, you can use the following command:

```
pswayam@pswayam-VirtualBox:~$ snort –dev –l ./log -h $HOME_
NET -c /etc/snort/ snort.conf
```

Please note that *snort.conf* is the name of your rule file and *HOME_NET* is a variable whose value is defined in the configuration file. The rule set defined in the *snort.conf* file will be applied to each packet, and a decision will be made on whether or not action needs to be taken. The default *snort. conf* references several other rule files, so it is a good idea to read through the entire *snort.conf* file before calling it from the command line.

It is also important to note here that if you are going to use Snort over a long period as an IDS, then do not use the *–v* switch in the command line for the sake of speed. When used as a network IDS, Snort provides near real-time intrusion detection capabilities.

Sometimes, you may be interested in the performance of Snort. In which case, you can use the *–b* and *–A fast* options. With these options, the packets will be logged in *tcpdump* format and you can expect very minimal alerts. The following command can be used for this purpose:

```
 pswayam@pswayam-VirtualBox:~$ snort –b –A fast –c/etc/snort/
snort.conf -lib
```

Snort applies its rules to packets in a specific order. The default order is that *Alert* rules will be applied first, then the *Pass* rules, and finally the *Log* rules. Snort provides a *–o* switch option to change the default rule application behaviour to *Pass* rules, then *Alert*, and then *Log*, as shown in the following command:

```
pswayam@pswayam-VirtualBox:~$ snort -d -h 192.168.1.0/24 -l
./log -c /etc/snort/snort.conf -o
```

Another important concept that we need to understand in Snort is output configurations. Snort provides a variety of options to display the output and the detected information. Please note that many Snort administrators use third-party applications to monitor and investigate the information generated by Snort. To do this, Snort must output the data in a particular format. One can enable multiple output plugins and these allow a variety of tools to be employed by Snort administrators. Let us have a quick look at some of these output configurations.

a. *Alert syslog:* UNIX based systems use the syslog facility to consolidate the generated messages. Snort-generated information can be presented to syslog in a number of ways. In this output configuration, the format looks like the following:

```
Output alert_syslog: <facility> <Priority>
```

An example is: *output alert_syslog: LOG_AUTH LOG ALERT*

b. *Log tcpdump:* This logs packets to the *tcpdump* file format. There are a variety of applications that can read this format.

c. *MySQL:* MySQL logging requires that the database support be present on the system running Snort, and that it is in the form of the MySQL client software and libraries.

After getting a clear understanding of Snort, a question will arise on where we need to place it. Putting Snort on each side of a firewall can be most enlightening. It is a best practice to run Snort behind a firewall, as this adds a layer of protection and is easier to manage.

Snort is considered the first choice when it comes to network IDS in many organisations. Cost-effectiveness and robustness are the two main parameters behind why people select it. But now with the several commercial versions of Snort available, industry support can be considered another reason behind its popularity. Added to this, the open source Snort community is also very active in providing the much needed support and enhancements of various features.

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